آغا خان یونیورسٹی ایگزامینیشن بورڈ

AGA KHAN UNIVERSITY EXAMINATION BOARD

Notes from E-Marking Centre on HSSC-II Chemistry Annual Examinations 2024

Introduction

This document has been produced for the teachers and candidates of Higher Secondary School Certificate (HSSC) Part II Chemistry. It contains comments on candidates' responses to the 2024 HSSC-II Examination indicating the quality of the responses and highlighting their relative strengths and weaknesses.

E-Marking Notes

This includes overall comments on candidates' performance on every question and *some* specific examples of candidates' responses that support the mentioned comments. Please note that the descriptive comments represent an overall perception of the better and weaker responses as gathered from the e-marking session. However, the candidates' responses shared in this document represent some specific example(s) of the mentioned comments.

Teachers and candidates should be aware that examiners may ask questions that address the Student Learning Outcomes (SLOs) in a manner that requires candidates to respond by integrating knowledge, understanding and application skills they have developed during the course of study. Candidates are advised to read and comprehend each question carefully before writing the response to fulfil the demand of the question.

Candidates need to be aware that the marks allocated to the questions are related to the answer space provided on the examination paper as a guide to the length of the required response. A longer response will not in itself lead to higher marks. Candidates need to be familiar with the command words in the SLOs which contain terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the command words. Words such as 'how', 'why' or 'what' may also be used.

General Observations

Overall, most candidates achieved success in constructing good responses that shows candidates' grasp of the subject and their ability to apply analytical skills to various types of questions. Some questions lacked comprehensive explanations, for example, reasons for importing chemicals, interpretation of IR spectrum and harmful effects of carbon dioxide on the environment. However, more candidates remained successful in constructing a good response specially in identification of functional groups, and radical mechanism reactions. Several responses were observed showing candidates' critical thinking abilities, for example, in Q.5 while mentioning reasons for importing chemicals, candidates gave a variety of responses including focusing on green chemistry and difficulties in mining in remote areas or areas where there are high-security risk factors. The concepts on which teachers need to focus and facilitate candidates in strengthening their concepts are mentioned below:

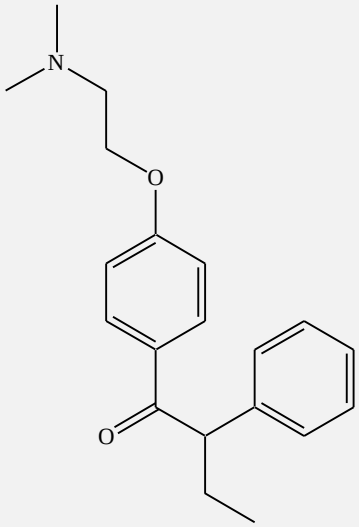
- Constructing chemical equations with an emphasis on balancing equations.
- Fulfiling the question's requirement(s) and including equations where necessary.

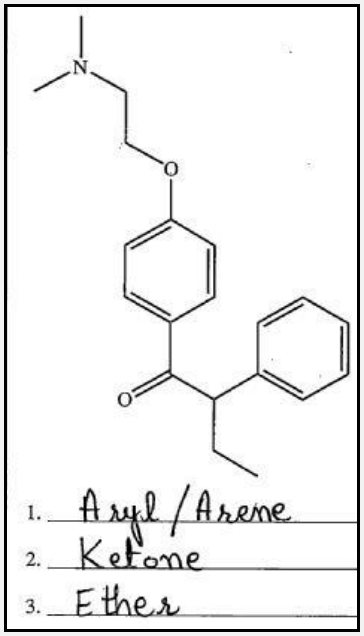
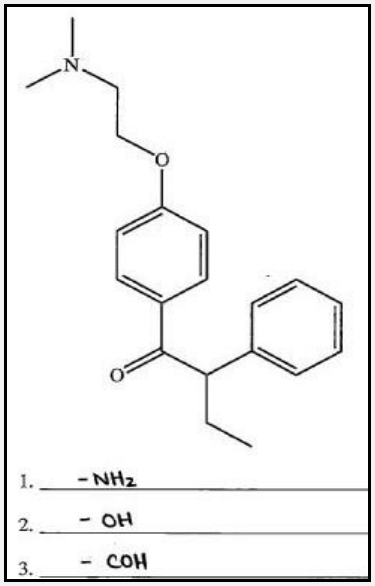
- Interpretation of spectrum and identification of common peaks.
- Rules for IUPAC nomenclature and identification of specific and multi-functional groups while naming.
- Identification of arene as a functional group rather than benzene or phenyl.
- Differentiation between primary, secondary and tertiary amine.
- Reasons for insolubility of alkyl halides in water.
- Comparative analysis of reactivity between carboxylic acid and its derivatives.

Note: Candidates responses shown in this report have not been corrected for grammar, spelling, format, or information.

DETAILED COMMENTS

Constructed Response Questions (CRQs)

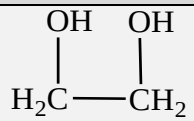
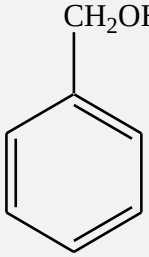
Question No. 1	
Question Text	Identify any THREE functional groups in the given structure of an organic molecule.  1. _____ 2. _____ 3. _____
SLO No.	15.2.2
SLO Text	Identify a molecule's functional group (i.e. alkane, alkene, alkyne, arene, halide, alcohol, ether, amine, nitrile, nitro, sulphide, sulphoxide, sulphone, thiol, aldehyde, ketone, carboxylic acid, ester, acid amide, acid chloride, acid anhydride).
Max Marks	3
Cognitive Level	U*
Checking Hints	1 mark for identifying each functional group (any THREE required)
Overall Performance	Generally, the question was well attempted. However, a few candidates depicted the need to understand the difference between the molecule's terminal methyl radical and terminal hydrogen. It was noted that many candidates circled the heteroatom in the provided molecular structure instead of identifying and writing down the correct functional group in

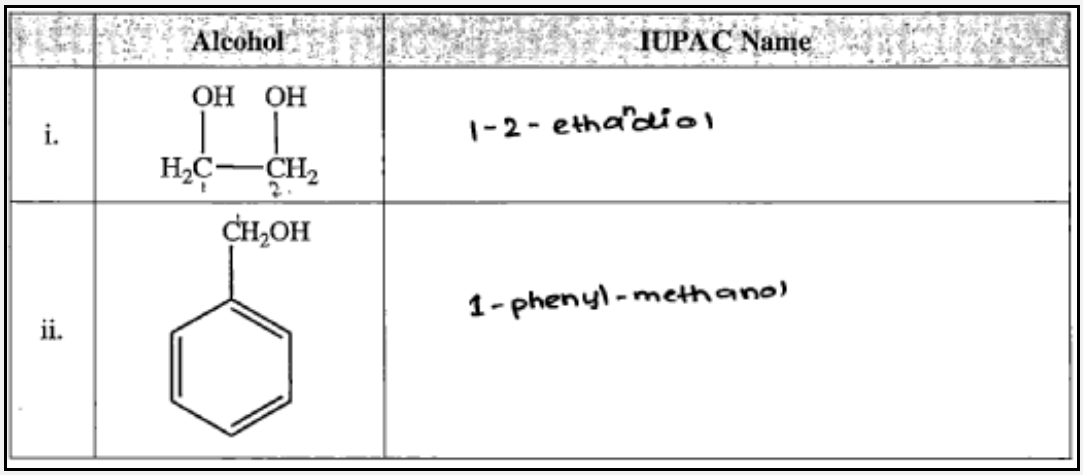
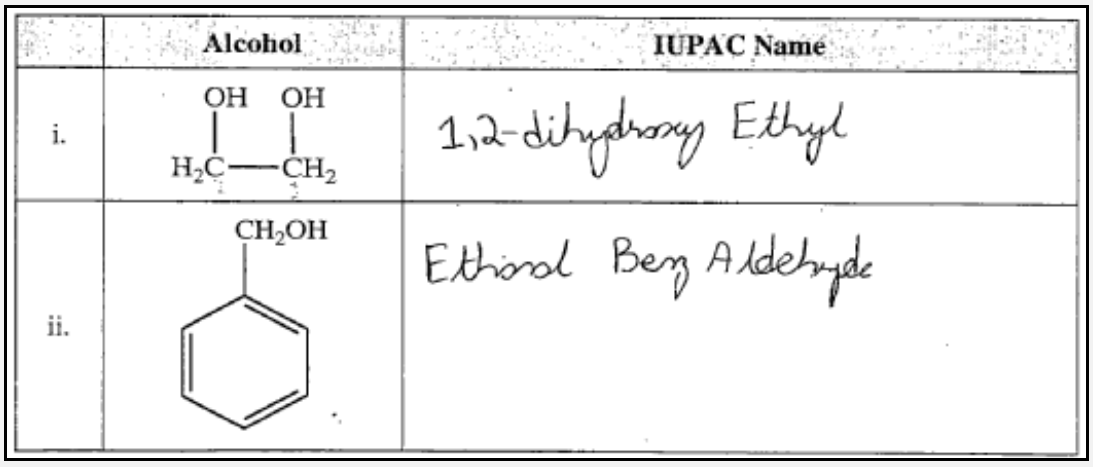
	the designated space. This indicates a lack of understanding of the command word, keyword, and what was being asked in the question.
Description of Better Responses	*Better responses* demonstrated of the correct identification of the functional groups and showed a strong grasp on the concept of heteroatoms. Candidates were able to recognise ketone as a carbonyl with two alkyl groups on each side of the carbon, a tertiary amine with nitrogen-bearing three alkyl groups, an ether as oxygen with two alkyl groups, and an arene (benzene).
Image of Better Response	
Description of Weaker Responses	Several minor errors were noticed in some of the *weaker responses*. For instance, some candidates incompletely identified functional groups, only mentioning carbonyl instead of the complete name or formula i.e., ketone. Additionally, a few candidates were unable to recognise arene (benzene) as a functional group, and some mistakenly identified a carboxylic group instead of a ketone. There were also instances where primary amine was incorrectly identified as tertiary amine, indicating a failure to recognise alkyl groups attached to the nitrogen. Lastly, a few responses encircled the functional group instead of writing the names and formulae.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy** Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration ** For description of each Pedagogy, refer to Annexure A	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login
Any Additional Suggestion:		

*K = Knowledge U = Understanding A = Application and other higher-order cognitive skills

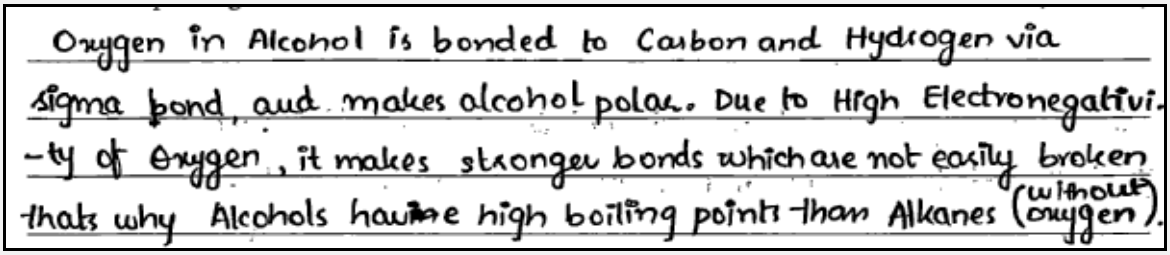
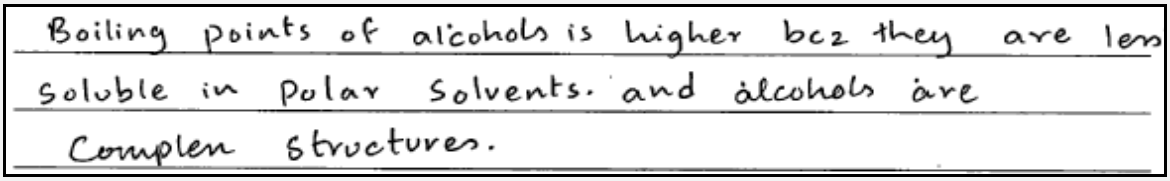
Question No. 2a	
Question Text	Write the names of the following alcohols as suggested by the International Union of Pure and Applied Chemistry (IUPAC). i.  ii. 
SLO No.	18.1.1
SLO Text	Apply IUPAC and trivial system for naming different alcohols.
Max Marks	2
Cognitive Level	A
Checking Hints	1 mark for each correct IUPAC name (2 required)
Overall Performance	Overall, the candidates' performance in this question concerning the simplest IUPAC nomenclature at the HSSC II level was unsatisfactory.
Description of Better Responses	*Better responses* demonstrated a clear understanding of the rules for naming the provided alcohol (alkyl and aryl) according to IUPAC. Most candidates were able to apply the

	IUPAC rules for the first structure. i.e., 1,2-ethandiol; however, a few responses correctly named the aryl alcohol i.e., phenyl methanol or 1-phenyl methanol.
Images of Better Responses	
Description of Weaker Responses	It was observed in *weaker responses,* that candidates struggled with IUPAC naming conventions. They made errors in identifying the number of carbons in the main chain and confused the phenyl group with benzene. Additionally, they incorrectly referred to methanol as benzene and used hydroxyl instead of the suffix '-ol' for alcohol. Some of the candidates also used common names instead of the correct IUPAC names such as dihydroxy ethane and benzyl alcohol. In other instances, candidates misidentified the benzene ring as the suffix instead of methanol. While discussing the second structure, the positions of the double bond were incorrectly labelled in the benzene ring. Furthermore, they also mistakenly used 'di' for two alcoholic groups in the same molecule, calling the first molecule 1,2-diethanol instead of the correct 1,2-ethanediol. Furthermore, many candidates incorrectly named the first molecule as 1,2-dihydroxyethanol, indicating a misunderstanding of using 'hydroxy' and the suffix '-ol' for alcohols.
Images of Weaker Responses	

Suggestions for improvement (Highlight all that apply)

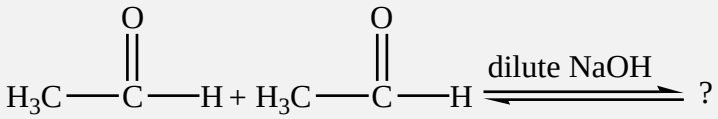
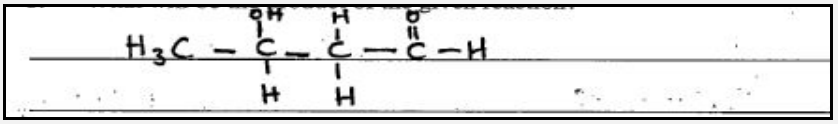
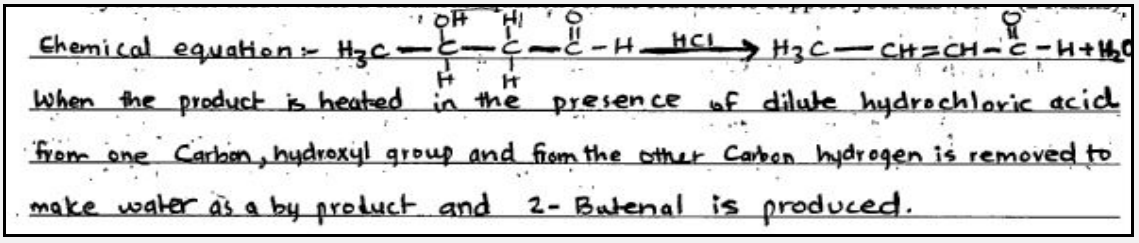
Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login
Any Additional Suggestion: Teachers should begin with the basics of functional groups and hydrocarbon structures, using analogies and mnemonics to simplify concepts. Engage students with interactive activities like molecular modelling and peer collaboration and reinforce learning through regular quizzes and real-world examples.		

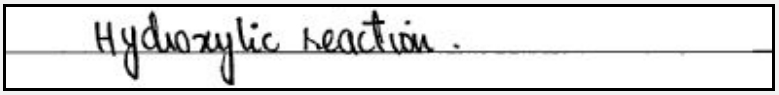
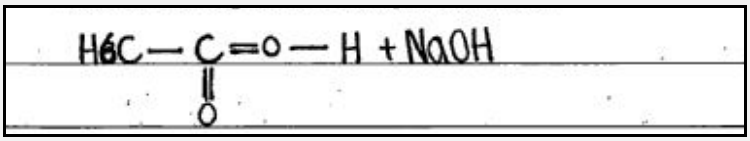
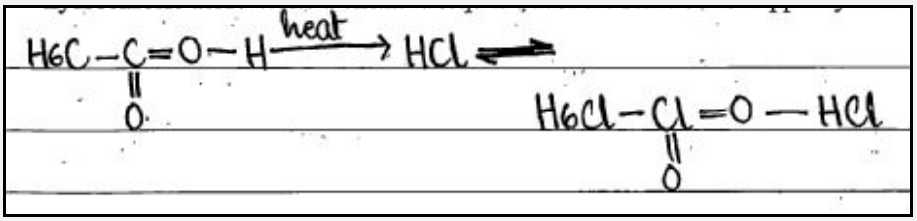
Question No. 2b	
Question Text	Briefly describe why the boiling points of alcohols are much higher than those of the corresponding alkanes.
SLO No.	18.1.2
SLO Text	Describe the physical properties and structure of alcohol.
Max Marks	2
Cognitive Level	U
Checking Hints	1 mark for writing about hydrogen bonding (absence or presence) 1 mark for writing about the high amount of energy required to break attractive forces/ hydrogen bonding
Overall Performance	Overall, the candidates' performance in this question demonstrated a solid understanding of the effect of weak bonding and the energy demand for boiling points in alcohols.
Description of Better Responses	*Better responses* demonstrated a clear understanding of how hydrogen bonding occurs among alcohol molecules because of the polarity of the -OH bond. Candidates also noted the high energy requirement to overcome the strong attractive forces in alcohol as compared to alkanes. Some candidates also mentioned the relationship between bond strength and bond energy when discussing alcohol.

Image of Better Response	
Description of Weaker Responses	Several *weaker responses* showed a lack of understanding of the attractive forces in alcohol and alkane. Some candidates discussed the solubility of polar solvents, which was not necessary for the question. A few responses successfully identified the attractive forces but did not explain the energy or heat required to overcome these forces. Some candidates also incorrectly mentioned hydrogen bonding, demonstrating a misconception of the basic concepts.
Image of Weaker Response	

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	**Preferred Pedagogy Used for this SLO**	**Assessment Strategies**
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Any Additional Suggestion:		

Question No. 3	
Question Text	Consider the given chemical reaction.  a. Identify the type of reaction shown above. b. What will be the product of the given reaction? c. What will happen if the product of the given reaction is heated in the presence of dilute hydrochloric acid? Write a chemical equation for the reaction to support your answer.
SLO No.	19.4.1
SLO Text	Describe the base catalysed nucleophilic addition reaction of aldehydes and ketones, i.e. a. addition of hydrogen cyanide b. addition of Grignard reagent c. addition of sodium bisulphate d. Aldol condensation e. Cannizzaro's reaction f. haloform (iodoform) reaction.
Max Marks	4
Cognitive Level	U
Checking Hints	a. 1 mark for correct identification b. 1 mark for showing the product c. 1 mark for showing loss of water 1 mark for showing an unsaturated carbonyl compound
Overall Performance	The overall performance in this question was satisfactory. Apart from the part of writing an equation in part 'c', candidates struggled to describe in-depth points related to Aldol condensation.
Description of Better Responses	*Better responses* demonstrated a solid understanding of chemical reactions and how to write equations. Many candidates correctly identified the aldol condensation as the reaction type in part 'a', and accurately depicted the product formula for the given reaction in part 'b'. These responses also correctly identified the product formed in part 'c' and demonstrated a clear understanding of how to create chemical equations.
Image of Better Response	Part 'a'  Part 'b'  Part 'c' 
Description of Weaker Responses	It was noted that in w*eaker responses*, that the aldol product was misidentified in part 'a' as a Cannizzaro reaction, hence preventing them from correctly addressing subsequent parts of the question. Candidates also lacked conceptual understanding of the dehydration

	concept in part 'c' of the question, where a water molecule was supposed to be released upon heating with HCl. Responses included incorrect chemical formulas in equations and a few responses showed the answer for part 'c' in part 'b', which could not be accepted as 'b' possessed completely different reaction conditions and reactants. A few responses scored points in part 'c' for water removal as a dehydration step, even though the main product was incorrect, indicating a lack of understanding of the overall concept.
Image of Weaker Response	Part 'a'  Part 'b'  Part 'c' 

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Pedagogy Used for that SLO	Assessment Strategies
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Any Additional Suggestion:		

Question No. 4	
Question Text	Describe the role of the following types of RNA in protein synthesis in a human cell. a. Messenger RNA b. Transfer RNA c. Ribosomal RNA
SLO No.	21.3.2
SLO Text	Describe the role of a. DNA in storing genetic information b. RNA in terms of protein synthesis.

Max Marks	3
Cognitive Level	U
Checking Hints	1 mark for mentioning ONE role of each type of RNA (THREE required)
Overall Performance	The overall performance on this question demonstrated a moderate to satisfactory level of understanding. Few candidates were unable to highlight the role of different types of RNA in protein synthesis.
Description of Better Responses	*Better response*s demonstrated the candidates' ability to accurately comprehend the role of RNA in protein synthesis. These responses also presented the roles of mRNA, tRNA, and rRNA. Candidates explained that mRNA transcribes genetic information from the gene to the cytoplasm. Furthermore, they identified tRNA's main role as decoding specific codons of mRNA during translation and carrying amino acids to the ribosome to assist in protein synthesis. The responses also correctly identified rRNA's role in assembling amino acids to synthesise proteins and explained that rRNA provides a site for protein synthesis in the cytoplasm.
Image of Better Response	a. Messenger RNA (1 Mark) Messenger RNA has a significant role and that is to "transcribe" a RNA from Chain DNA fragment and then "translate" it in Ribosomes to "give the sequence" of amino acids for protein synthesis b. Transfer RNA (1 Mark) Transfer RNA "brings the amino acid" to the ribosomes (rRNA) for the protein formation. Transfer RNA has anticodon with which it matches the amino acid sequence on (MRNA) c. Ribosomal RNA (1 Mark) ribosomal RNA is there to provide the "site" for protein synthesis". These synthesis occur in cytoplasm of cell and rRNA provides a site for synthesis on ribosomes.
Description of Weaker Responses	Some responses demonstrated a lack of understanding regarding the role of RNA in protein synthesis, incorrectly identifying RNA's function. Several responses mistakenly attributed the production of ribosomes to the Golgi apparatus. Additionally, some candidates inaccurately described energy transfer as occurring from the brain to cells. There was also confusion between the processes of transcription and translation, with some describing protein synthesis as a function of tRNA.
Image of Weaker Response	a. Messenger RNA (1 Mark) collect. msg from the whole body and sends to brain b. Transfer RNA (1 Mark) The brain transfer the msg to the particular body part through this c. Ribosomal RNA (1 Mark) work as a shield. and protect the body against harmful bacterias.

Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestion:		

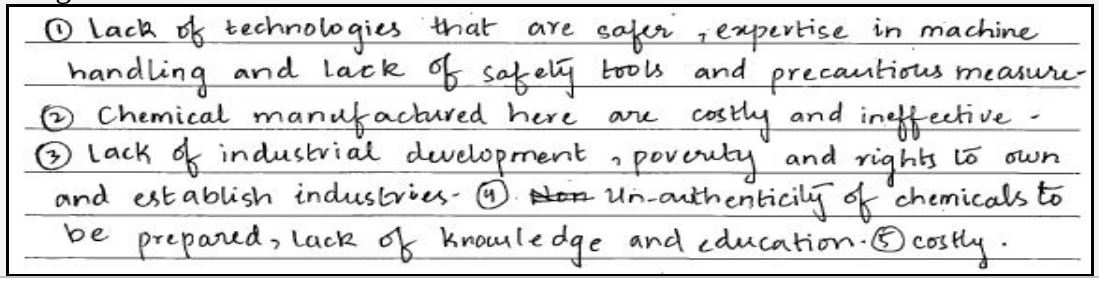
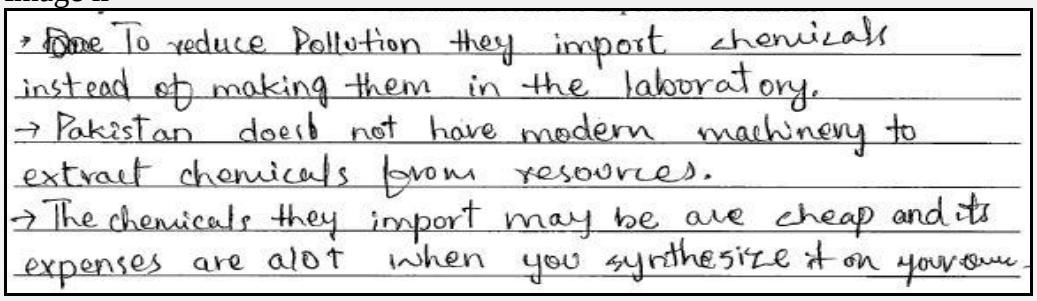
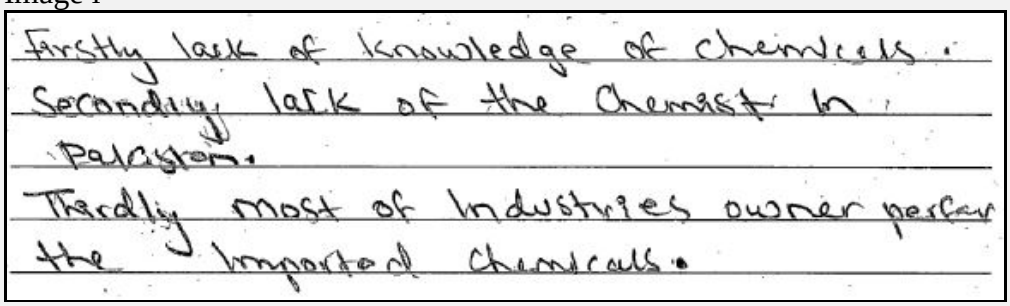
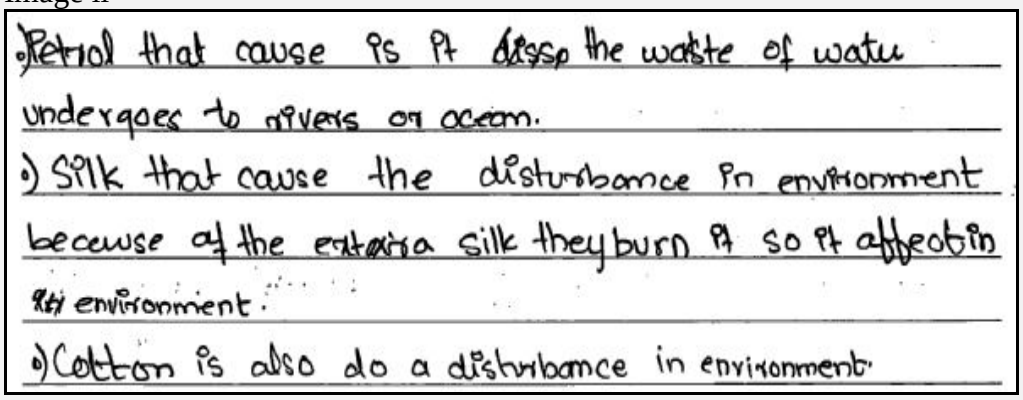
Question No. 5	
Question Text	In Pakistan, chemical industries are considered to have enough resources to manufacture chemicals independently; however, they prefer to import chemicals from foreign countries. Write any THREE factors that cause Pakistani industries to import these chemicals.
SLO No.	22.1.1
SLO Text	Discuss the importance of chemical industries for the economy of Pakistan.
Max Marks	3
Cognitive Level	U*
Checking Hints	1 mark for each factor (any THREE required)
Overall Performance	Overall, candidates performed well in this question given its thought-provoking nature. Many candidates focused on less common factors that were still relevant to the question.
Description of Better Responses	*Better responses* demonstrated a clear identification of factors and effectively utilised knowledge and understanding of environmental concerns, risk factors, and supply and demand related to chemical manufacturing. These responses highlighted the cost factor as paramount, along with the lack of modern technology and equipment in Pakistan. Candidates also discussed the environmental risks associated with chemicals and emphasised the importance of maintaining a green environment. Concerns were raised about the purity of synthesised chemicals, with a preference for imported chemicals due to their perceived higher quality. Additionally, responses noted that importing chemicals saves time and energy, making it a more efficient solution to meet high demand.
Images of Better Responses	Image i  ① Lack of technologies that are safer, expertise in machine handling and lack of safety tools and precautious measure- ② Chemical manufactured here are costly and ineffective - ③ Lack of industrial development, poverty and rights to own and establish industries- ④ ~~Non~~ Un-authenticity of chemicals to be prepared, lack of knowledge and education- ⑤ costly.

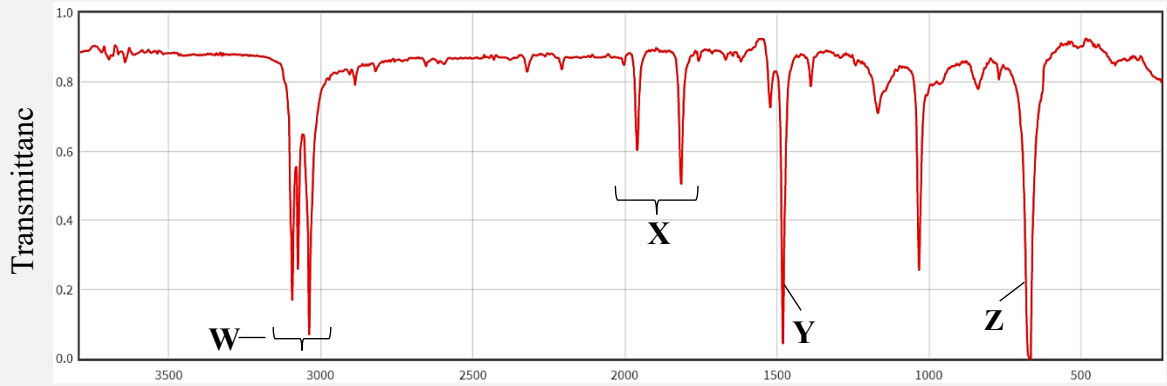
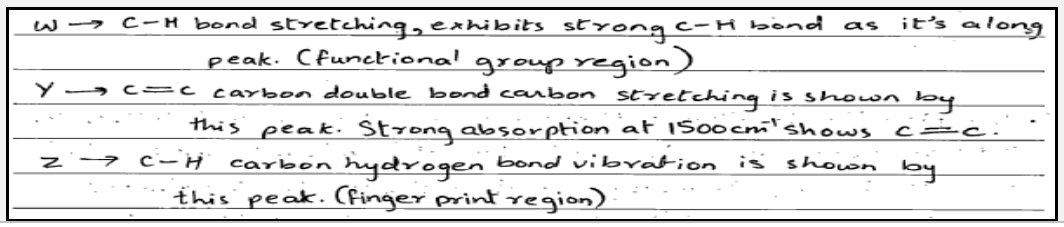
	Image ii  • Done To reduce Pollution they import chemicals instead of making them in the laboratory. → Pakistan doesn't not have modern machinery to extract chemicals from resources. → The chemicals they import may be are cheap and its expenses are alot when you synthesize it on yourown.
Description of Weaker Responses	*Weaker responses* demonstrated a lack of comprehension. Many answers were vague and irrelevant, often repeating the same points. A number of candidates erroneously mentioned the lack of industries in Pakistan, which is not factual. Additionally, some candidates incorrectly stated that Pakistan lacks skilled personnel capable of manufacturing chemicals, which is also not a valid point. Several responses incorrectly cited a lack of resources in Pakistan, despite the question specifying that industries are assumed to have sufficient resources.
Images of Weaker Responses	Image i  Firstly lack of knowledge of chemicals. Secondly lack of the Chemists in Pakistan. Thirdly most of Industries owner prefer the Imported Chemicals. Image ii  •Petrol that cause Ps Pt disso the waste of water undergoes to rivers or ocean. •) Silk that cause the disturbance Pn environment because of the extensa silk they burn Pt so Pt affectin Pts environment. •)Cotton is also do a distribance in environment.

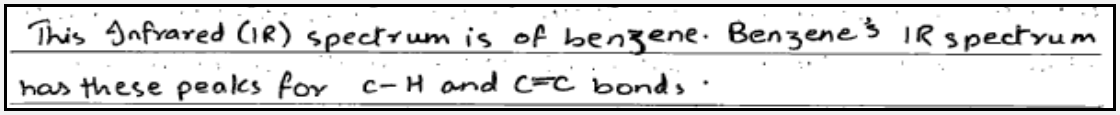
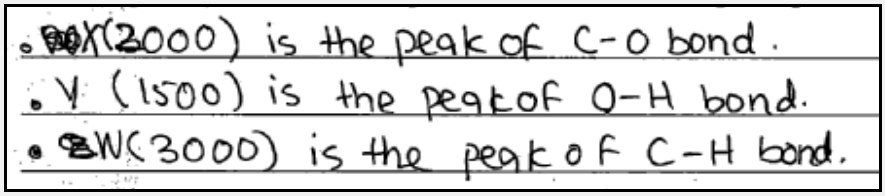
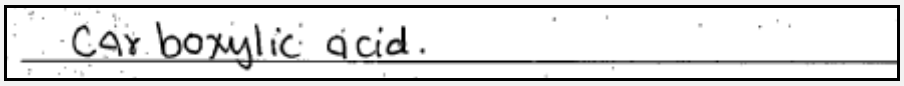
Suggestions for improvement (Highlight all that apply)

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• Utilise the resource guide for additional materials	Practical Demonstration	

Any Additional Suggestion:

Question No. 6	
Question Text	Consider the given Infrared (IR) spectrum of an organic compound.  Wavenumber (cm^{-1}) Source: https://webbook.nist.gov/cgi/inchi?ID=C71432&Type=IR-SPEC&Index=2 a. Interpret any THREE labelled peaks (**W**, **X**, **Y** or **Z**) in the given spectrum. b. Based on the interpretation in part a, identify the compound.
SLO No.	24.1.8
SLO Text	Interpret the infrared (IR) spectra of benzene, acetone, acetic acid and ethanol.
Max Marks	4
Cognitive Level	A
Checking Hints	a. 1 mark for interpreting each labelled peak (any 3 required) 1 mark will be awarded if the candidate writes any one point of each labelled peak. b. 1 mark for identification of benzene
Overall Performance	The overall performance on this question was unsatisfactory. Notably, some candidates identified the compound as benzene without interpreting specific wavenumbers on the spectrum, indicating they relied on memorising the shapes of the four spectra given in the SLO rather than demonstrating a true understanding of IR spectrum analysis.
Description of Better Responses	*Better responses* demonstrated a thorough understanding of interpreting the IR spectrum. These answers accurately identified peaks **W**, **X**, **Y**, and **Z** within their respective ranges (W: 3200-2800 cm^{-1}, X: 2000-1700 cm^{-1}, Y: 1478 cm^{-1}, Z: 674 cm^{-1}). Additionally, candidates correctly identified benzene in part 'b' based on the interpretation of the labelled peaks. In *better responses*, candidates correctly identified the **W** peak as a C-H stretch.
Image of Better Response	Part 'a'  W → C–H bond stretching, exhibits strong C–H bond as it's along peak. (Functional group region) Y → C=C carbon double bond carbon stretching is shown by this peak. Strong absorption at 1500cm^{-1} shows C=C. Z → C–H carbon hydrogen bond vibration is shown by this peak. (finger print region)

	Part 'b' 
Description of Weaker Responses	*Weaker responses* showed a limited understanding of interpreting the given spectrum and failed to identify the correct compound. Instead, they only mentioned the ranges of the labelled peaks. Common errors included incorrectly identifying the **Y** peak as a C=O group, misplacing the **Z** peak in the fingerprint region, converting transmittance to absorbance, and incorrectly labelling the saturated C-H stretches as falling above the region of 3200 cm^{-1}. Many candidates were unable to identify the **Z** peak as a C-H bend. A few candidates also misidentified the **X** peak as an unsaturated bond rather than an aromatic ring.
Image of Weaker Response	Part 'a'  Part 'b' 

Suggestions for improvement (Highlight all that apply)

Maximising SLO Achievement	Preferred Pedagogy Used for this SLO	Assessment Strategies
• Identify the expectation of command words (use Command Word Guide) • Ensure the content is taught at the relevant cognitive level • Identify necessary content required (skills + concepts) • Review past paper questions on the concept • Utilise the resource guide for additional materials	• Story Board • Cause and Effect • Fish and Bone • Concept Mapping • Audio Visual Resources • Think, Pair and Share • Knowledge Platform videos • Questioning Technique (Socratic approach) • Practical Demonstration	• Past paper questions • Discussion on E-Marking Notes • AKU-EB Digital Learning Solution powered by Knowledge Platform https://akueb.knowledgeplatform.com/login
Any Additional Suggestion:		

Extended Response Questions (ERQs)

Extended response questions offered a choice between parts 'a' and 'b'

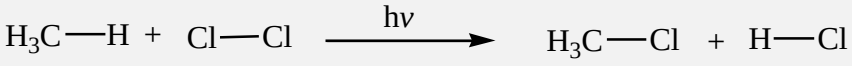
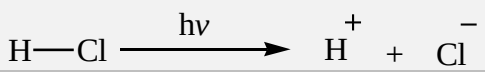
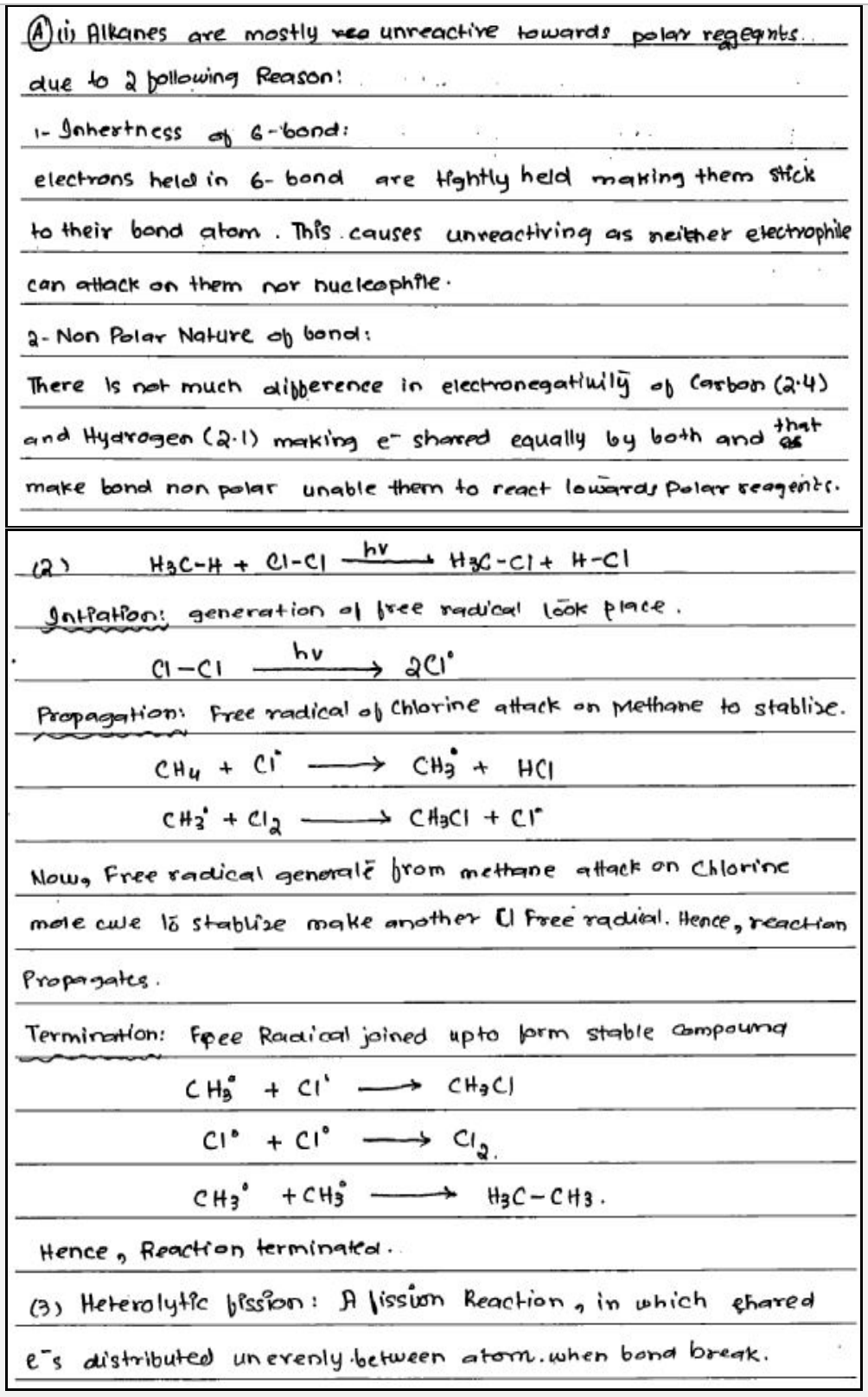
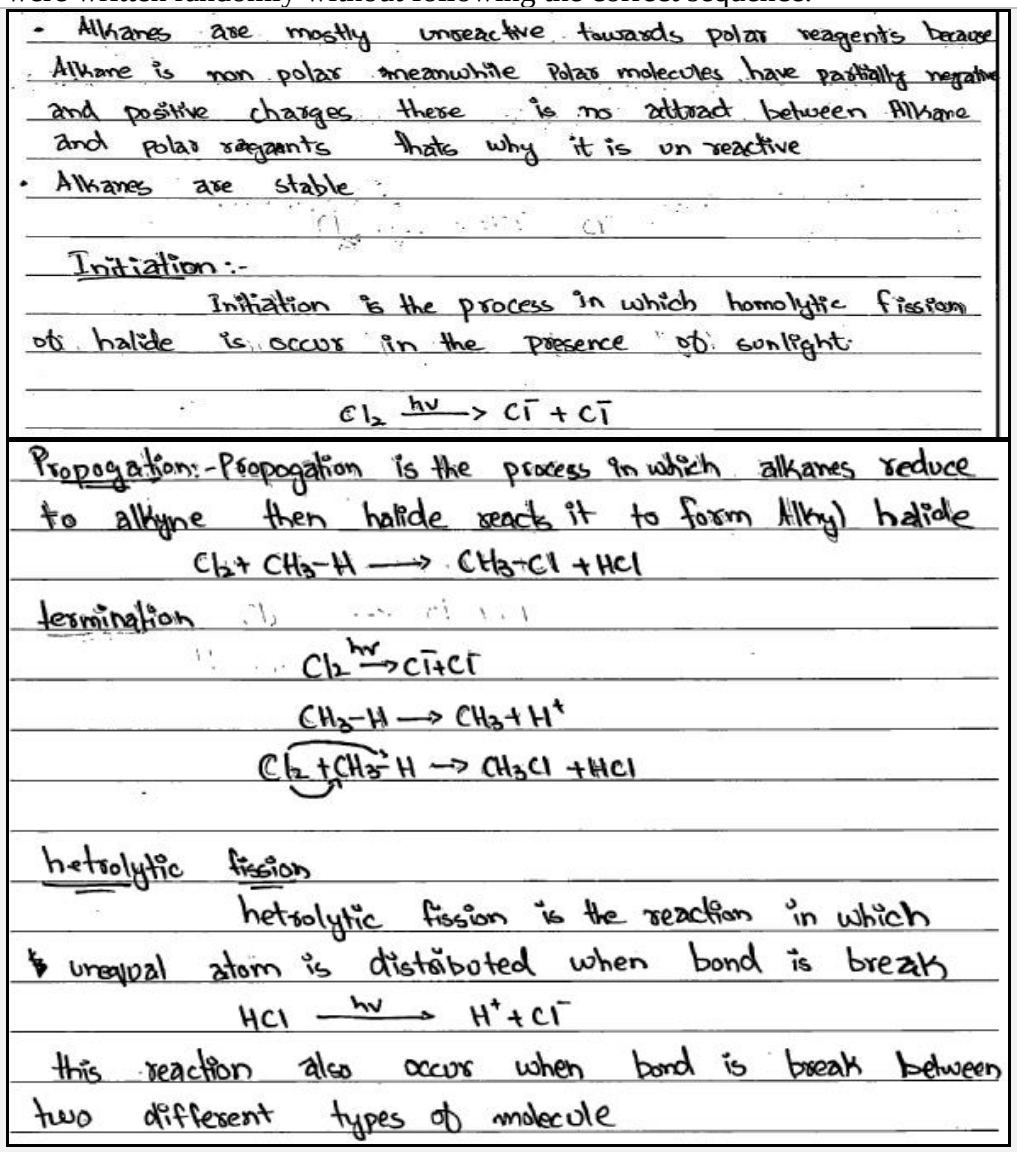
Question No. 7a	
Question Text	i. Give TWO reasons why alkanes are mostly unreactive towards polar reagents. ii. Write the steps of the free radical mechanism by showing the initiation, propagation and termination steps for the given reaction.  iii. Identify the type of fission reaction in the given chemical equation. 
SLO No.	16.2.1; 16.2.2
SLO Text	Explain unreactive nature of alkanes towards polar reagents. Explain homolytic and heterolytic fission, free radical initiation, propagation and termination.
Max Marks	7
Cognitive Level	U
Checking Hints	i. 1 mark for each reason (2 required) ii. 1 mark for showing the initiation step 1 mark for showing each propagation step (2 required) 1 mark for showing a termination step iii. 1 mark for each correct identification
Overall Performance	Candidates generally performed well on this question and elaborated all the steps of the radical chain reaction.
Description of Better Responses	In part 'i', *better responses* clearly explained the reasons for the unreactive nature of alkanes. Candidates understood that two marks required two distinct reasons. In part 'ii', candidates demonstrated a solid understanding of chain reaction mechanisms, securing full marks by correctly identifying the reactants and products in the equations. These responses indicated a strong grasp of the concept of chain reactions, correctly showing radical formation in equations by identifying one molecule, a radical as reactants, and a radical and molecule as products. Candidates extended the equation using the same reaction strategy, accurately identifying the initiation, propagation, and termination steps. Responses included correct radical signs in equations and presented the equations in the correct sequence. Some responses included more than the required two propagation steps and one termination step, but they still correctly demonstrated the process. Additionally, candidates successfully identified heterolytic fission as a type of reaction, showing an understanding of the difference between the cleavage of two identical atoms and two different atoms.

Image of Better Response	 (A) (i) Alkanes are mostly unreactive towards polar reagents due to 2 following Reason: 1- Inhertness of σ-bond: electrons held in σ- bond are tightly held making them stick to their bond atom. This causes unreactiving as neither electrophile can attack on them nor nucleophile. 2- Non Polar Nature of bond: There is not much difference in electronegatiwily of Carbon (2.4) and Hydrogen (2.1) making e⁻ shared equally by both and that as make bond non polar unable them to react towards Polar reagents. (2) $H_3C-H + Cl-Cl \xrightarrow{h\nu} H_3C-Cl + H-Cl$ Initiation: generation of free radical took place. $Cl-Cl \xrightarrow{h\nu} 2Cl^{\cdot}$ Propagation: Free radical of Chlorine attack on Methane to stablize. $CH_4 + Cl^{\cdot} \longrightarrow CH_3^{\cdot} + HCl$ $CH_3^{\cdot} + Cl_2 \longrightarrow CH_3Cl + Cl^{\cdot}$ Now, Free radical generate from methane attack on Chlorine make cwe to stablize make another Cl Free radical. Hence, reaction Propagates. Termination: Free Radical joined upto form stable compound $CH_3^{\cdot} + Cl^{\cdot} \longrightarrow CH_3Cl$ $Cl^{\cdot} + Cl^{\cdot} \longrightarrow Cl_2$. $CH_3^{\cdot} + CH_3^{\cdot} \longrightarrow H_3C-CH_3$. Hence, Reaction terminated. (3) Heterolytic fission: A fission Reaction, in which shared e⁻s distributed unevenly between atom. when bond break.
Description of Weaker Responses	*Weaker responses* exhibited an unclear understanding of the rubrics, leading to lost marks. In part 'i', candidates often provided only one reason for the non-reactivity of alkanes or repeated the same reason with different wording. In part 'ii', many candidates omitted the radical sign in equations or mistakenly used charges instead. Some responses incorrectly depicted the initiation step by showing positive and negative ions instead of two radicals. These responses revealed a lack of understanding of the radical mechanism, presenting complete equations rather than illustrating radical formation, indicating the question was not fully read. Some candidates showed two reactants as radicals without producing a

	radical in the product. Additionally, there was a failure to follow the rubric, with some responses showing only one propagation step, thus losing a mark. In several cases, equations were written randomly without following the correct sequence.
Image of Weaker Response	

- Alkanes are mostly unreactive towards polar reagents because Alkane is non polar meanwhile Polar molecules have partially negative and positive charges there is no attract between Alkane and polar reagents thats why it is un reactive
- Alkanes are stable

Initiation :-

Initiation is the process in which homolytic fission of halide is occur in the presence of sunlight

$Cl_2 \xrightarrow{h\nu} C\bar{l} + C\bar{l}$

Propogation:- Propogation is the process in which alkanes reduce to alkyne then halide reacts it to form Alkyl halide

$Cl_2 + CH_3-H \longrightarrow CH_3-Cl + HCl$

termination

$Cl_2 \xrightarrow{hv} C\bar{l}+C\bar{l}$

$CH_3-H \longrightarrow CH_3 + H^+$

$Cl_2 + CH_3^- H \longrightarrow CH_3Cl + HCl$

hetrolytic fission

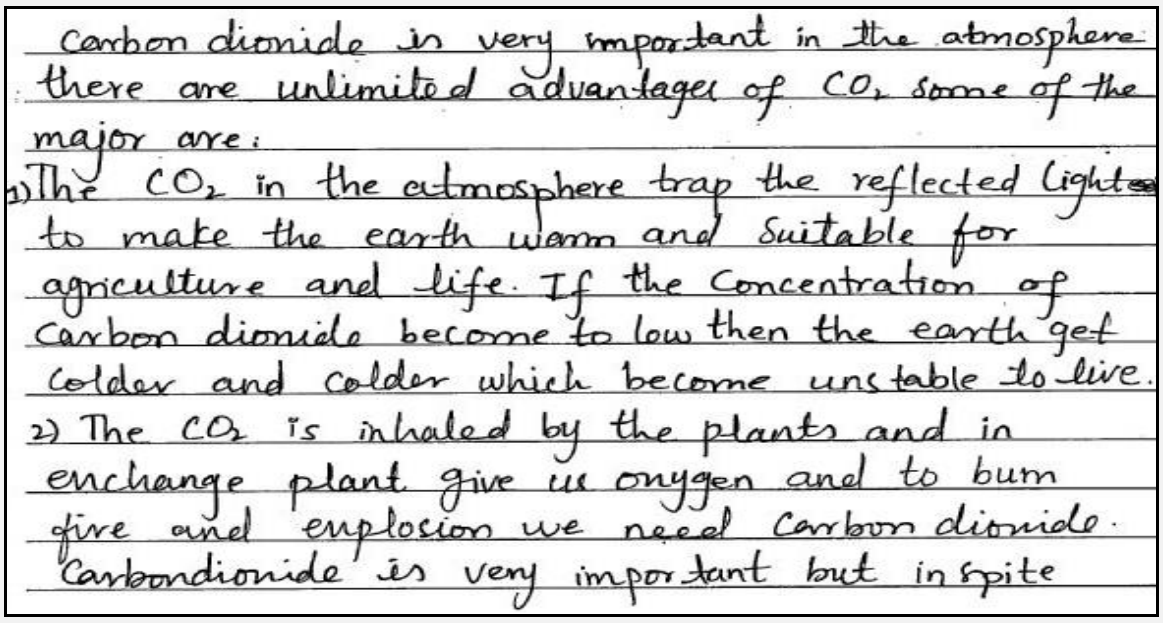
hetrolytic fission is the reaction in which unequal atom is distributed when bond is breaks

$HCl \xrightarrow{h\nu} H^+ + Cl^-$

this reaction also occur when bond is break between two different types of molecule

Suggestions for improvement (Highlight all that apply)

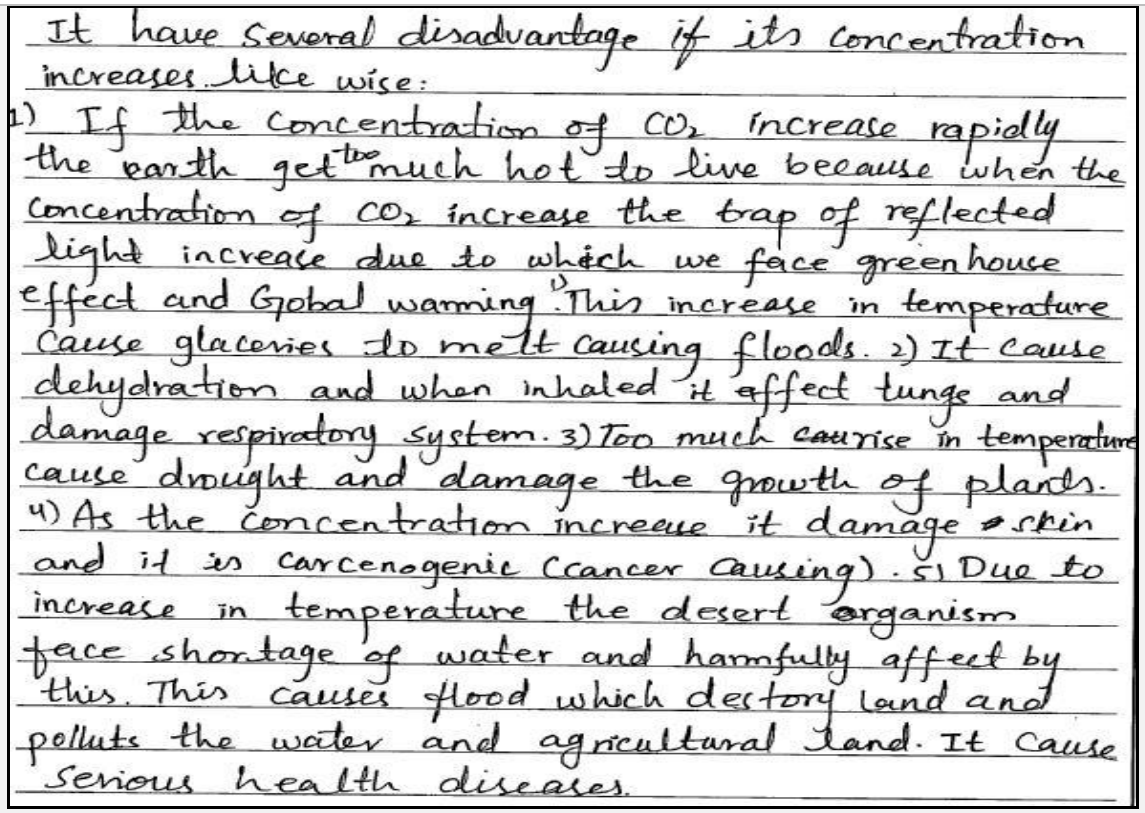
Maximising SLO Achievement	**Preferred Pedagogy Used for this SLO**	**Assessment Strategies**
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Any Additional Suggestion:		

Question No. 7b	
Question Text	i. Describe the importance of carbon dioxide gas in the atmosphere. ii. Write any FIVE harmful effects of the rise in concentration of carbon dioxide on the environment.
SLO No.	23.1.7
SLO Text	Explain climate change as a result of greenhouse effect and global warming.
Max Marks	7
Cognitive Level	U
Checking Hints	i. 1 mark for describing each role of carbon dioxide (any 2 required) ii. 1 mark for each correct impact (any 5 required)
Overall Performance	Overall, the candidates' responses demonstrated both the advantageous aspects of CO_2 presence and the environmental issues caused by the excessive production of this greenhouse gas. However, a few candidates incorrectly associated CO_2 with human and physiological detrimental impacts rather than limiting their discussion to environmental effects.
Description of Better Responses	*Better responses*, in part 'i', included a precise and necessary explanation of the importance of CO_2 in the atmosphere. In part 'ii', candidates successfully identified the correct harmful effects of the rise in CO_2 concentration. They demonstrated a clear understanding of how CO_2 contributes to global warming and the consequences of global warming. Despite the open-ended question, students secured marks based on their basic understanding of environmental chemistry, showing competent knowledge of the subject matter.
Image of Better Response	

Carbon dioxide is very important in the atmosphere there are unlimited advantages of CO_2 some of the major are:

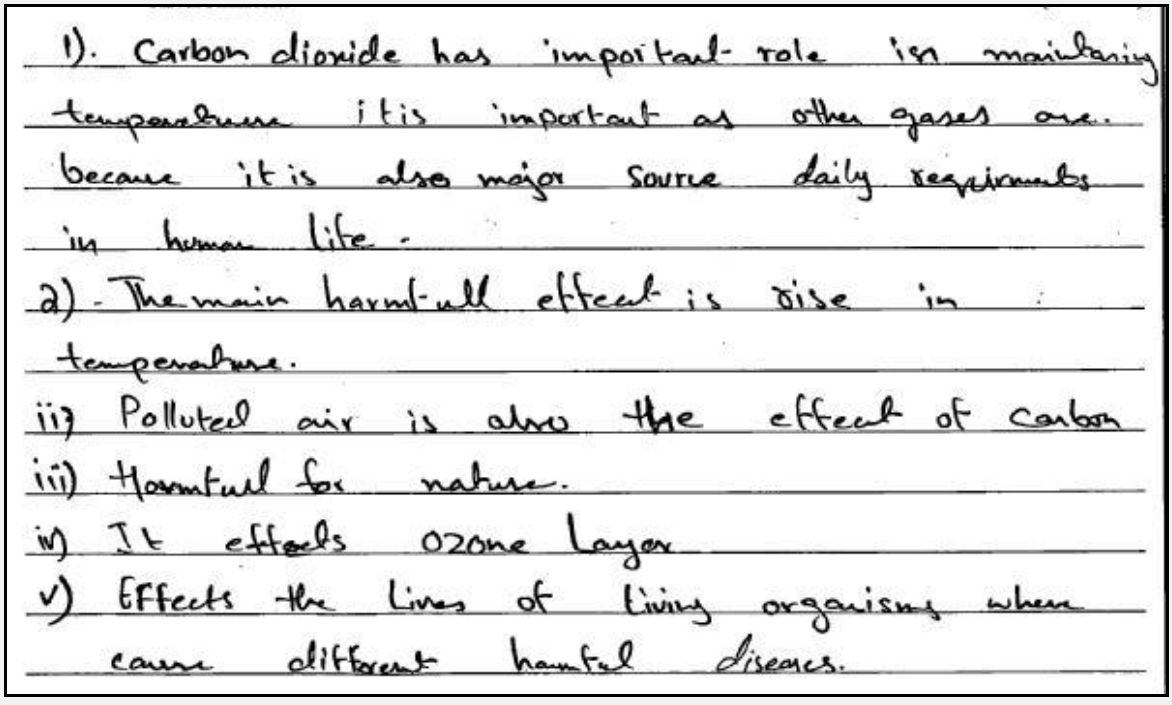
1) The CO_2 in the atmosphere trap the reflected lights to make the earth warm and suitable for agriculture and life. If the concentration of carbon dioxide become to low then the earth get colder and colder which become unstable to live.

2) The CO_2 is inhaled by the plants and in exchange plant give us oxygen and to burn fire and explosion we need carbon dioxide. Carbondioxide is very important but in spite



It have several disadvantage if its concentration increases. like wise:

1) If the concentration of CO_2 increase rapidly the earth get too much hot to live because when the concentration of CO_2 increase the trap of reflected light increase due to which we face greenhouse effect and Gobal warming. This increase in temperature cause glaceries to melt causing floods. 2) It cause dehydration and when inhaled it affect lungs and damage respiratory system. 3) Too much caurise in temperature cause drought and damage the growth of plants. 4) As the concentration increase it damage skin and it is carcenogenic (cancer causing). 5) Due to increase in temperature the desert organism face shortage of water and harmfully affect by this. This causes flood which destory land and polluts the water and agricultural land. It cause serious health diseases.

Description of Weaker Responses	A considerable number of candidates left the question unanswered, indicating a lack of basic environmental knowledge. In part 'i', *weaker responses* demonstrated an unclear understanding of the rubrics, losing marks by providing only one point instead of two, and failed to identify CO_2 as a greenhouse gas. Many candidates also did not mention CO_2's role in the photosynthesis process. In part 'ii', many responses included irrelevant points and repeated the same facts in different wording. Some candidates incorrectly stated that CO_2 is responsible for ozone depletion and a major contributor to acid rain. These responses failed to link CO_2's role to global warming and its consequences, such as melting glaciers and rising sea levels.
Image of Weaker Response	

1). Carbon dioxide has 'importaut- role 'in maintaing temperature it is 'importaut as other gases are. becaue it is also major source daily requirments in human life.

2) - The main harmfull effect is rise in temperature.

ii) Polluted air is also the effect of carbon

iii) Harmfull for nature.

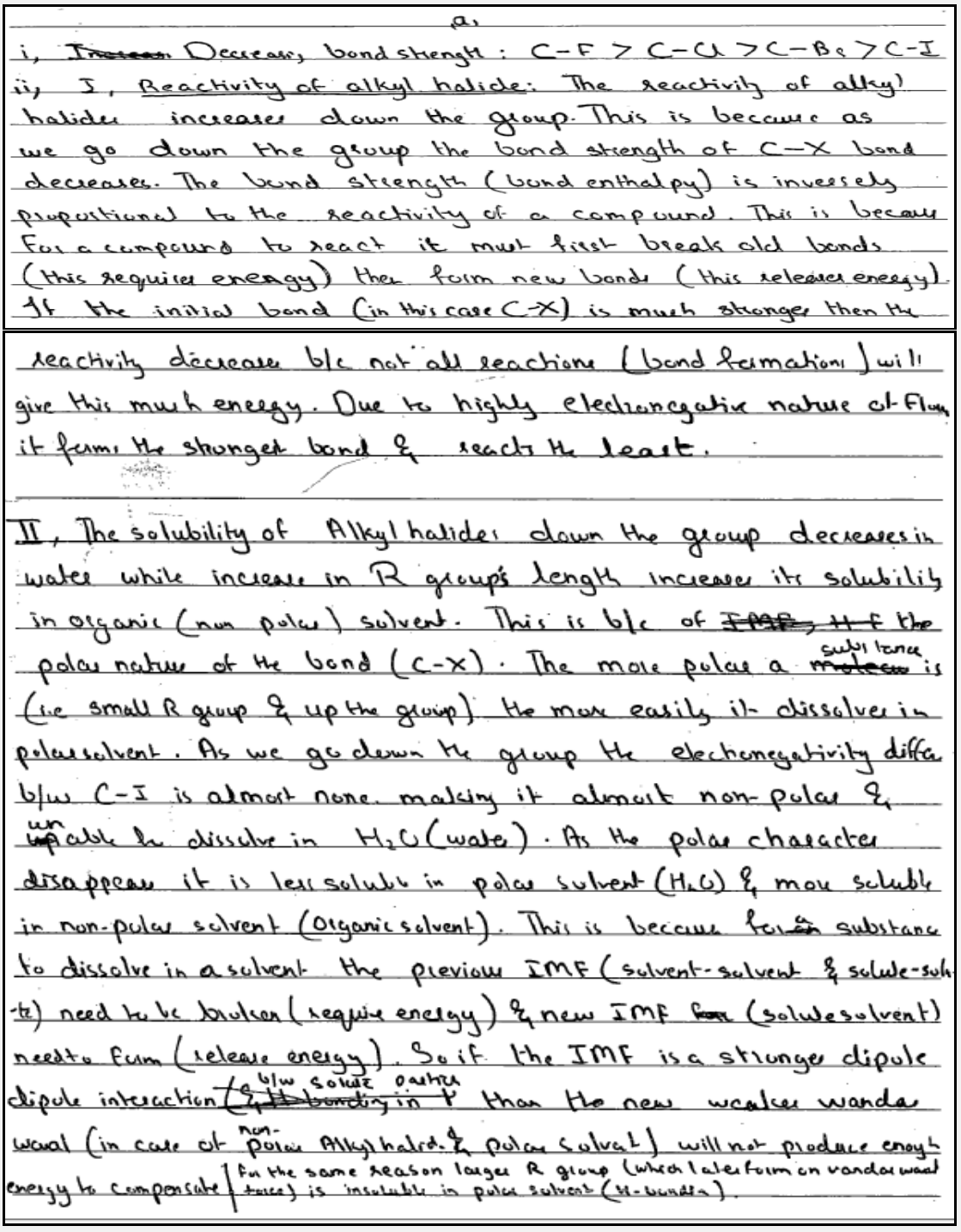
iv) It effects ozone Layer

v) Effects the Lives of living orgaisms when cause different hamful diseases.

Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestion:		

Question No. 8a	
Question Text	i. Write the order of bond strength in carbon halides, starting from the strongest to the weakest. ii. Based on the order of bond strength mentioned in part i, describe the following: I. Reactivity of alkyl halides II. Solubility of alkyl halides in water and organic solvents
SLO No.	17.1.2
SLO Text	Discuss physical properties and reactivity of different alkyl halides on the basis of bond energy.
Max Marks	7
Cognitive Level	U
Checking Hints	i. 1 mark for writing the correct order of carbon halide bonds ii. I. 1 mark for describing the reactivity of alkyl halides 1 mark for connecting it to the order described in part i. II. 2 marks for describing the solubility of alkyl halides in water 1 mark for writing about the low solubility in water 1 mark for correct reasoning based on bond energy/ strength 2 marks for describing the solubility of alkyl halides in organic solvents. 1 mark for writing about the high solubility in organic solvents 1 mark for correct reasoning based on bond energy/ strength
Overall Performance	Overall, candidates showed satisfactory performance in this question. Most candidates were able to correctly identify the other reactivity of carbon halide bonds, while others confused it with hydrogen halides.
Description of Better Responses	*Better responses* demonstrated an in-depth understanding of bond strength, reactivity, and solubility of alkyl halides. In part 'i', candidates correctly identified the order of bond strength of carbon-halide bonds, both in increasing and decreasing order. In part 'ii' (I), they explained the reactivity of alkyl halides based on bond strength and bond length, accurately describing iodoalkanes as the most reactive and fluoroalkanes as the least

	reactive. These responses also provided a well-supported explanation of the solubility of alkyl halides in water. They noted that the attractive forces between halogenoalkanes and water molecules are not strong enough to disrupt the hydrogen bonds in water, resulting in low solubility. Additionally, candidates explained that the larger size of the halogen atom contributes to decreased solubility down the group due to reduced bond strength. Some responses further elaborated on polyhalogenated alkanes having greater density than water, causing them to settle at the bottom, and included explanations related to polarity and electronegativity differences. In part 'ii' (II), candidates accurately described the ease of solubility of alkyl halides in organic solvents such as CCl_4 and ether, attributed to the 'like dissolves like' phenomenon.
Image of Better Response	
Description of Weaker Responses	Candidates provided incorrect orders of bond strength in part 'i', with some erroneously explaining the bond strength of hydrogen halides instead of carbon-halide bonds. *Weaker responses* in part 'ii' (I), demonstrated an unclear understanding of the relationship between bond strength and reactivity. These responses incorrectly stated that reactivity is directly

	proportional to bond strength/ bond energy and inversely proportional to bond length, and incorrectly ranked reactivity in terms of primary, secondary, and tertiary alkyl halides. Some responses incorrectly attributed the differences in solubility to electronegativity differences affecting polarity. In part 'ii' (II), candidates misunderstood the concept of solubility, linking it to the alkyl chain length and incorrectly asserting that increased alkyl chain length decreases solubility. They also misrelated solubility to reactivity, mistakenly suggesting that R-I is the most soluble alkyl halide.
Image of Weaker Response	(a) i) The order of bond strength in carbon halides is:- C-I > C-Cl > C-Br > C-F This is the bond strength order because flourine is the most reactive whereas Iodine is the least reactive. The C-F bond can easily cleave upon the attacking of a new atom, but the C-I does not cleave eqsily due to less reactivity and high bond strength. ii) I) Reactivity of Alkyl Halides:- As discussed, since the bond strength of C-I is high and C-F is low, the C-I bond is less reactive and the C-F bond is more reactive. The order of Reactivity is:- C-F > C-Br > C-Cl > C-I II) Solubility of Alkyl Halides in Water and Organic Solvents:- The solubility of Alkyl halides in aqueous medium would be high. The solubility of alkyl halides order would be same as the order of reactivity. The order of Solubility:- C-F > C-Br > C-Cl > C-I

Suggestions for improvement (Highlight all that apply)

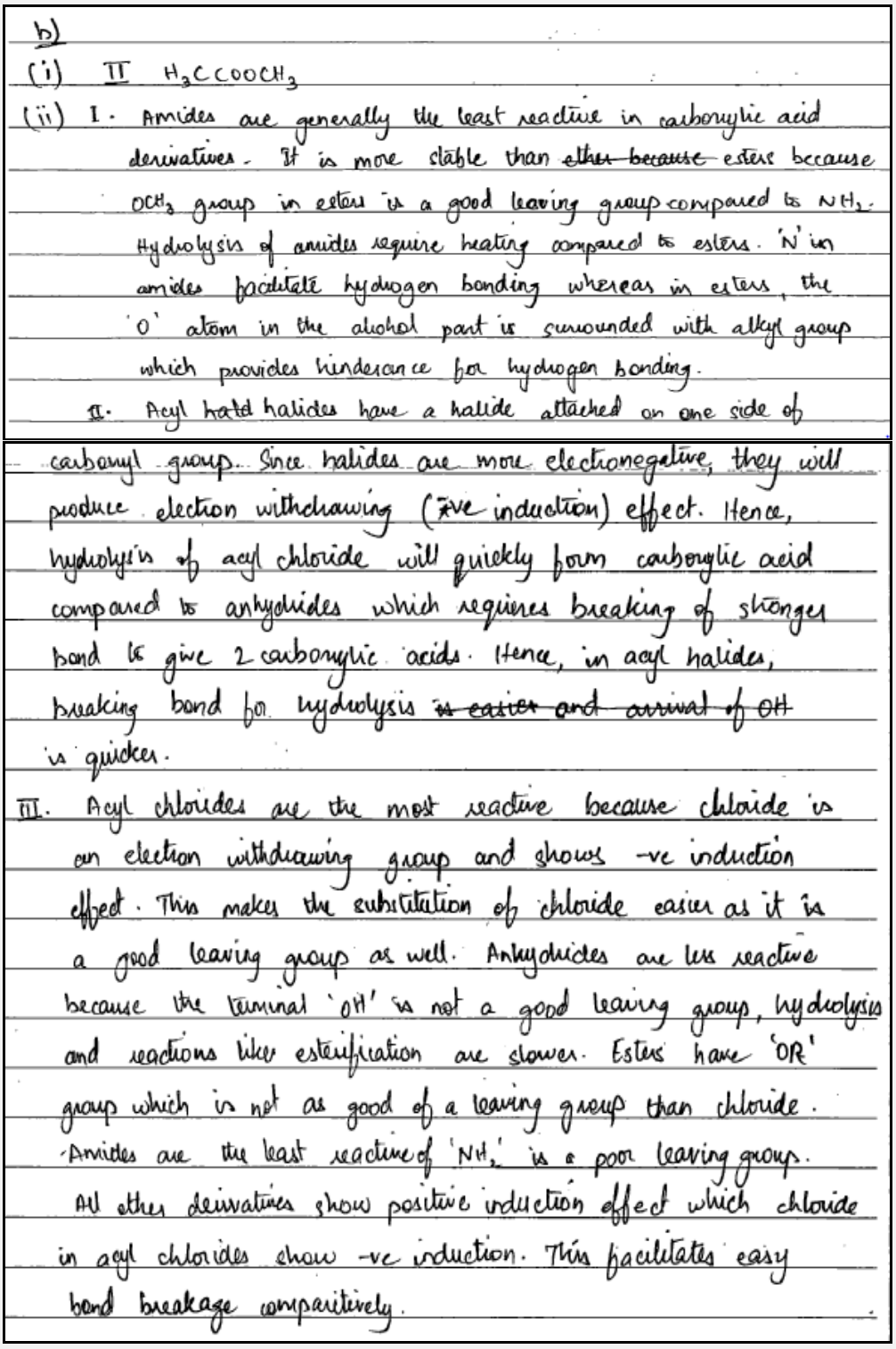
Maximising SLO Achievement	Pedagogy Used for that SLO	Assessment Strategies
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Any Additional Suggestion:		

Question No. 8b	
Question Text	i. Identify which of the following readily undergoes nucleophilic substitution reaction. I. H_3CCOOH II. $H_3CCOOCH_3$ ii. Describe reasons for each of the following statements. I. Amides are more stable than esters. II. Hydrolysis of acyl halides occurs faster than anhydrides. III. Acyl chloride is the most reactive among the derivatives of carboxylic acid.
SLO No.	20.5.2
SLO Text	Compare the reactivity of different derivatives of carboxylic acid (i.e. acyl halides, acid anhydrides, esters and amides).
Max Marks	7
Cognitive Level	U*
Checking Hints	a. 1 mark for correct identification of ester b. 2 marks for describing reasons for each statement (3 required)
Overall Performance	Overall, the candidates showed below satisfactory performance. The responses depicted a lack of an in-depth understanding of the comparative reasoning of carboxylic acid derivatives.
Description of Better Responses	*Better responses* in part 'i' demonstrated a strong grasp of the reactivity of carboxylic acid derivatives, identifying $H_3CCOOCH_3$ as the most reactive to undergo nucleophilic substitution reactions. In part 'ii' (I), candidates explained that nitrogen's low electronegativity allows resonance due to the lone pair of electrons on the nitrogen atom. They effectively illustrated the concept of resonance with electron-donating groups by drawing appropriate resonating structures. Some candidates also noted that esters exhibit the least stability because resonance does not occur due to the similar environment of carbon concerning C-O and C=O bonds. In part 'ii' (II), candidates discussed the hydrolysis of alkyl halides in terms of leaving groups. They noted that halides in acyl halides are good leaving groups and more acidic as compared

to acetate, which is a moderate leaving group. Responses also highlighted the candidates' understanding about the carbonyl group in acyl halides as more electrophilic due to the inductive effect of halides. Some candidates also explained the less electrophilic character of carbonyl carbon in anhydrides due to electron donation by the anhydride oxygen atom.

In part 'ii' (III), candidates explicitly explained why acyl chlorides are the most reactive among carboxylic acid derivatives. They highlighted that the electronegative chlorine acts as an electron-withdrawing group, making the carbonyl carbon more electrophilic and facilitating nucleophilic attack. They also mentioned that Cl^- is a better-leaving group, which enhances reactivity. Candidates also described amides as the most unreactive due to the poor leaving group ability of the nitrogen atom.

Images of Better Responses



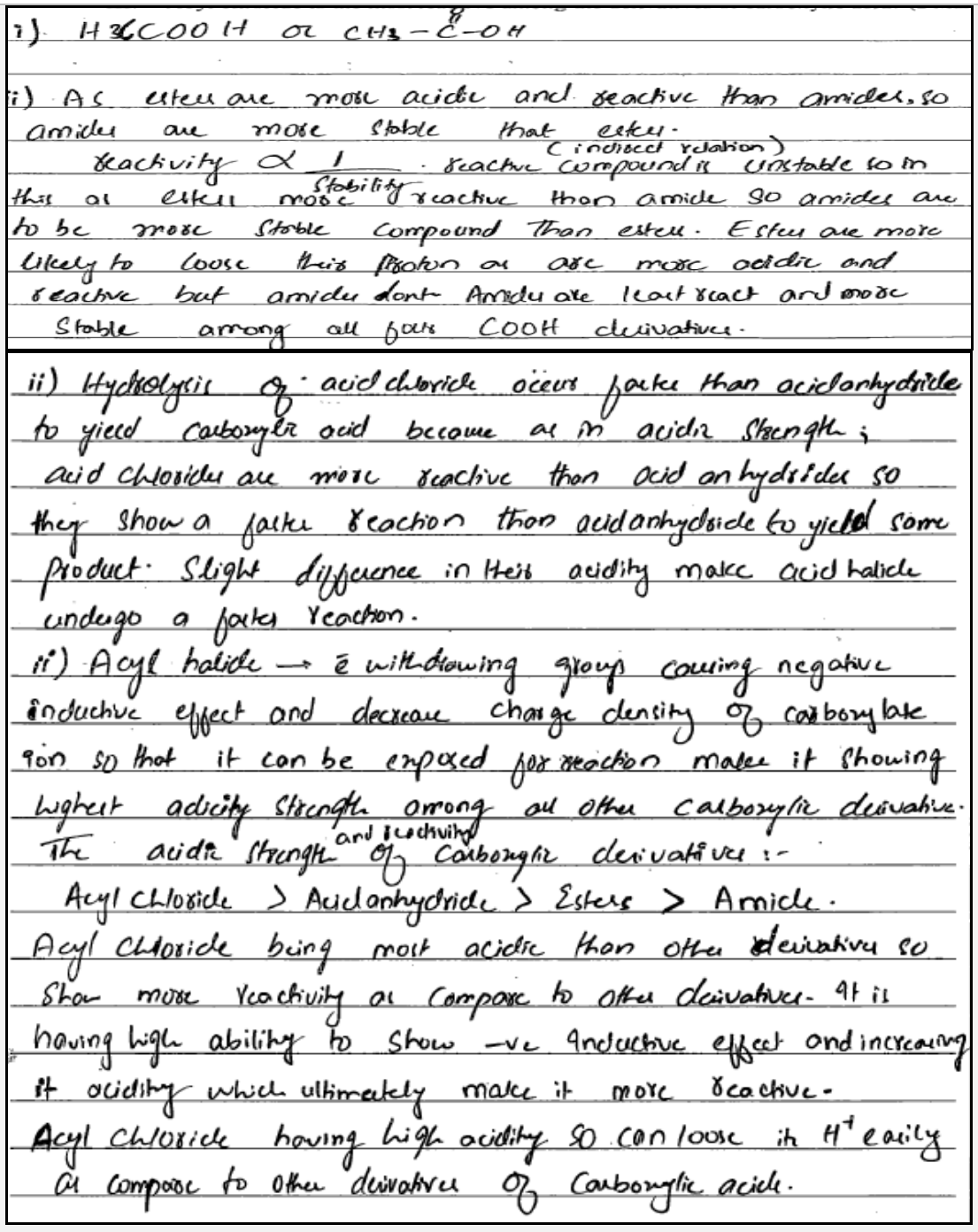
b)

(i) II $H_3CCOOCH_3$

(ii) I. Amides are generally the least reactive in carboxylic acid derivatives. It is more stable than ~~ether because~~ esters because OCH_3 group in esters is a good leaving group compared to NH_2. Hydrolysis of amides require heating compared to esters. 'N' in amides facilitate hydrogen bonding whereas in esters, the 'O' atom in the alcohol part is surrounded with alkyl group which provides hinderance for hydrogen bonding.

II. Acyl halides have a halide attached on one side of carbonyl group. Since halides are more electronegative, they will produce election withdrawing (-ve induction) effect. Hence, hydrolysis of acyl chloride will quickly form carboxylic acid compared to anhydrides which requires breaking of stronger bond to give 2 carboxylic acids. Hence, in acyl halides, breaking bond for hydrolysis ~~is easier and arrival of OH~~ is quicker.

III. Acyl chlorides are the most reactive because chloride is an election withdrawing group and shows -ve induction effect. This makes the substitution of chloride easier as it is a good leaving groups as well. Anhydrides are less reactive because the terminal 'OH' is not a good leaving group, hydrolysis and reactions like esterification are slower. Esters have 'OR' group which is not as good of a leaving group than chloride. Amides are the least reactive of 'NH_2' is a poor leaving groups. All other deirvatives show positive induction effect which chloride in acyl chlorides show -ve induction. This facilitates easy bond breakage comparitively.

Description of Weaker Responses	Weaker responses in part 'i' contained errors in identifying $H_3CCOOCH_3$ as the most reactive compound for nucleophilic substitution reactions. In part 'ii' (I), candidates incorrectly explained the relationship between reactivity and stability, relating it to acidity. They also described the basic nature of the amide group without adequately connecting it to resonance or stability. In part 'ii' (II), candidates failed to explain the concept of leaving groups and the inductive effect of the Cl group. They incorrectly based their explanations of hydrolysis on bond strength and bond energy rather than on the leaving group's ability and inductive effects. In part 'ii' (III), responses acknowledged the halide group as the most electronegative element, but candidates failed to elaborate on its effect on the electrophilicity of the carbonyl group, leading to incomplete or incorrect explanations of the reactivity of carboxylic acid derivatives.
Image of Weaker Response	 i) H_3CCOOH or $CH_3-\overset{O}{\overset{\|}{C}}-OH$ ii) As ester are more acidic and reactive than amides. so amides are more stable that ester. Reactivity $\propto \frac{1}{\text{Stability}}$ (indirect relation). reactive compound is unstable so in this as ester more reactive than amide so amides are to be more stable compound than ester. Esters are more likely to loose their proton as are more acidic and reactive but amides don't. Amides are least react and more stable among all four COOH derivatives. ii) Hydrolysis of acid chloride occurs faster than acidanhydride to yield carboxylic acid because of its acidic strength; acid chlorides are more reactive than acid anhydrides so they show a faster reaction than acidanhydride to yield same product. Slight difference in their acidity make acid halide undergo a faster reaction. ii) Acyl halide → e withdrawing group causing negative inductive effect and decrease charge density of carboxylate ion so that it can be exposed for reaction makes it showing highest acidity strength among all other carboxylic derivative. The acidic strength and reactivity of carboxylic derivatives :- Acyl Chloride > Acidanhydride > Esters > Amide. Acyl Chloride being most acidic than other derivatives so show more reactivity as compare to other derivatives. It is having high ability to show -ve inductive effect and increasing it acidity which ultimately make it more reactive. Acyl Chloride having high acidity so can loose its H^+ easily as compare to other derivatives of Carboxylic acide.

Suggestions for improvement (Highlight all that apply)

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Any Additional Suggestion:		

Annexure A: Pedagogies Used for Teaching the SLOs

Pedagogy: Storyboard
Description: A visual pedagogy that uses a series of illustrated panels to present a narrative, encouraging creativity and critical thinking. It helps learners organise ideas, sequence events, and comprehend complex concepts through storytelling.
Example: In a Literature class, students are tasked with creating storyboards to visually retell a novel. They draw key scenes, write captions, and present their stories to the class, enhancing their reading comprehension and fostering their imagination.

Pedagogy: Cause and Effect
Description: This pedagogy explores the relationships between actions and consequences. By analysing cause-and-effect relationships, learners develop a deeper understanding of how events are interconnected and how one action can lead to various outcomes.
Example: In a History class, students study the causes and effects of the Industrial Revolution. They research and discuss how technological advancements in manufacturing led to significant societal changes, such as urbanisation and labour reform movements.

Pedagogy: Fish and Bone
Description: A method that breaks down complex topics into main ideas (the fish) and supporting details (the bones). This visual approach enhances comprehension by highlighting essential concepts and their relevant explanations.
Example: During a Biology class on human anatomy, the teacher uses the fish and bone technique to teach about the human skeletal system. Teacher presents the main components of the human skeleton (fish) and elaborates on each bone's structure and function (bones).

Pedagogy: Concept Mapping
Description: An effective way to visually represent relationships between ideas. Learners create diagrams connecting key concepts, aiding in understanding the overall structure of a subject and fostering retention.
Example: In a Psychology assignment, students use concept mapping to explore the various theories of personality. They interlink different theories, such as Freud's psychoanalysis, Jung's analytical psychology, and Bandura's social-cognitive theory, to see how they relate to each other.

Pedagogy: Audio Visual Resources
Description: Incorporating multimedia elements like videos, images, and audio into lessons. This approach caters to different learning styles, making educational content more engaging and memorable.
Example: In a General Science class, the teacher uses a documentary-style video to teach about the solar system. The video includes stunning visual animations of the planets, interviews with astronomers, and background music, enhancing students' interest and understanding of space.

Pedagogy: Think, Pair, and Share
Description: A collaborative learning technique where students ponder a question or problem individually, then discuss their thoughts in pairs or small groups before sharing with the entire class. It fosters active participation, communication skills, and diverse perspectives.
Example: In a Literature in English class, the teacher poses a thought-provoking question about a novel's moral dilemma. Students first reflect individually, then pair up to exchange their opinions, and finally participate in a lively class discussion to explore different viewpoints.

Pedagogy: Questioning Technique (Socratic Approach)
Description: Based on Socratic dialogue, this method stimulates critical thinking by posing thought-provoking questions. It encourages learners to explore ideas, justify their reasoning, and discover knowledge through a process of inquiry.
Example: In an Ethics class, the instructor uses the Socratic approach to lead a discussion on the meaning of justice. By asking a series of probing questions, the students engage in a deeper exploration of ethical principles and societal values.

Pedagogy: Practical Demonstration
Description: A hands-on approach where learners observe real-life applications of theories or skills. Practical demonstrations enhance comprehension, skill acquisition, and problem-solving abilities by bridging theoretical concepts with real-world scenarios.
Example: In a Food and Nutrition class, the instructor demonstrates the proper technique for filleting a fish. Students observe and then practice the skill themselves, learning the practical application of knife skills and culinary precision.

(**Note:** The examples provided in this annexure serve as illustrations of various pedagogies. It is important to understand that these pedagogies are versatile and can be applied across subjects in numerous ways. Feel free to adapt and explore these techniques creatively to enhance learning outcomes in your specific context.)

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